

```

<!--
{
  "availability" : [
    "iOS: 2.0.0 -",
    "iPadOS: 2.0.0 -",
    "macCatalyst: 13.1.0 -",
    "macOS: 10.0.0 -",
    "tvOS: 9.0.0 -",
    "visionOS: 1.0.0 -",
    "watchOS: 1.0.0 -"
  ],
  "documentType" : "symbol",
  "framework" : "ObjectiveC",
  "identifier" : "/documentation/ObjectiveC/NSObject-swift.class",
  "metadataVersion" : "0.1.0",
  "role" : "Class",
  "symbol" : {
    "kind" : "Class",
    "modules" : [
      "Objective-C Runtime"
    ],
    "preciseIdentifier" : "c:objc(cs)NSObject"
  },
  "title" : "NSObject"
}
-->

```

NSObject

The root class of most Objective-C class hierarchies, from which subclasses inherit a basic interface to the runtime system and the ability to behave as Objective-C objects.

```

...

```

```

class NSObject
{
}

```

Topics

Initializing a Class

```
[`+ initialize`](/documentation/ObjectiveC/NSObject-swift.class/initialize())
```

Initializes the class before it receives its first message.

```
[`+ load`](/documentation/ObjectiveC/NSObject-swift.class/load())
```

Invoked whenever a class or category is added to the Objective-C runtime; implement this method to perform class-specific behavior upon loading.

Creating, Copying, and Deallocating Objects

```
[`+ alloc`](/documentation/ObjectiveC/NSObject-swift.class/alloc)
```

Returns a new instance of the receiving class.

```
[`+ allocWithZone:`](/documentation/ObjectiveC/NSObject-swift.class/allocWithZone:)

```

Returns a new instance of the receiving class.

```
[`- init`](/documentation/ObjectiveC/NSObject-swift.class/init())

```

Implemented by subclasses to initialize a new object (the receiver) immediately after memory for it has been allocated.

[`- copy`](/documentation/ObjectiveC/NSObject-swift.class/copy())

Returns the object returned by `copy(with:)`.

[`+ copyWithZone:`](/documentation/ObjectiveC/NSObject-swift.class/copyWithZone:)

Returns the receiver.

[`- mutableCopy`](/documentation/ObjectiveC/NSObject-swift.class/mutableCopy())

Returns the object returned by `mutableCopy(with:)` where the zone is `nil`.

[`+ mutableCopyWithZone:`](/documentation/ObjectiveC/NSObject-swift.class/mutableCopyWithZone:)

Returns the receiver.

[`- dealloc`](/documentation/ObjectiveC/NSObject-swift.class/dealloc)

Deallocates the memory occupied by the receiver.

[`+ new`](/documentation/ObjectiveC/NSObject-swift.class/new)

Allocates a new instance of the receiving class, sends it an [`.init()`](/documentation/ObjectiveC/NSObject-swift.class/init()) message, and returns the initialized object.

Identifying Classes

[`+ class`](/documentation/ObjectiveC/NSObject-swift.class/class)

Returns the class object.

[`+ superclass`](/documentation/ObjectiveC/NSObject-swift.class/superclass())

Returns the class object for the receiver's superclass.

[`+ isKindOfClass:`](/documentation/ObjectiveC/NSObject-swift.class/isKindOfClass(of:))

Returns a Boolean value that indicates whether the receiving class is a subclass of, or identical to, a given class.

Testing Class Functionality

[`+ instancesRespondToSelector:`](/documentation/ObjectiveC/NSObject-swift.class/instancesRespond(to:))

Returns a Boolean value that indicates whether instances of the receiver are capable of responding to a given selector.

Testing Protocol Conformance

[`+ conformsToProtocol:`](/documentation/ObjectiveC/NSObject-swift.class/conforms(to:))

Returns a Boolean value that indicates whether the target conforms to a given protocol.

otocol.

Obtaining Information About Methods

```
[`- methodForSelector:`](/documentation/ObjectiveC/NSObject-swift.class/methodFor:))
```

Locates and returns the address of the receiver's implementation of a method so it can be called as a function.

```
[`+ instanceMethodForSelector:`](/documentation/ObjectiveC/NSObject-swift.class/instanceMethod(for:))
```

Locates and returns the address of the implementation of the instance method identified by a given selector.

```
[`+ instanceMethodSignatureForSelector:`](/documentation/ObjectiveC/NSObject-swift.class/instanceMethodSignatureForSelector:)
```

Returns an `NSMethodSignature` object that contains a description of the instance method identified by a given selector.

```
[`- methodSignatureForSelector:`](/documentation/ObjectiveC/NSObject-swift.class/methodSignatureForSelector:)
```

Returns an `NSMethodSignature` object that contains a description of the method identified by a given selector.

Describing Objects

```
[`+ description`](/documentation/ObjectiveC/NSObject-swift.class/description())
```

Returns a string that represents the contents of the receiving class.

Supporting Discardable Content

```
[`autoContentAccessingProxy`](/documentation/ObjectiveC/NSObject-swift.class/autoContentAccessingProxy)
```

A proxy for the receiving object

Sending Messages

```
[`- performSelector:withObject:afterDelay:`](/documentation/ObjectiveC/NSObject-swift.class/perform(_:with:afterDelay:))
```

Invokes a method of the receiver on the current thread using the default mode after a delay.

```
[`- performSelector:withObject:afterDelay:inModes:`](/documentation/ObjectiveC/NSObject-swift.class/perform(_:with:afterDelay:inModes:))
```

Invokes a method of the receiver on the current thread using the specified modes after a delay.

```
[`- performSelectorOnMainThread:withObject:waitUntilDone:`](/documentation/ObjectiveC/NSObject-swift.class/performSelector(onMainThread:with:waitUntilDone:))
```

Invokes a method of the receiver on the main thread using the default mode.

```
[`- performSelectorOnMainThread:withObject:waitUntilDone:modes:`](/documentation/ObjectiveC/NSObject-swift.class/performSelector(onMainThread:with:waitUntilDone:modes:))
```

```
e:modes:))
```

Invokes a method of the receiver on the main thread using the specified modes.

```
[`- performSelector:onThread:withObject:waitUntilDone:`](/documentation/ObjectiveC/NSObject-swift.class/perform(_:on:with:waitUntilDone:))
```

Invokes a method of the receiver on the specified thread using the default mode.

```
[`- performSelector:onThread:withObject:waitUntilDone:modes:`](/documentation/ObjectiveC/NSObject-swift.class/perform(_:on:with:waitUntilDone:modes:))
```

Invokes a method of the receiver on the specified thread using the specified modes.

```
[`- performSelectorInBackground:withObject:`](/documentation/ObjectiveC/NSObject-swift.class/performSelector(inBackground:with:))
```

Invokes a method of the receiver on a new background thread.

```
[`+ cancelPreviousPerformRequestsWithTarget:`](/documentation/ObjectiveC/NSObject-swift.class/cancelPreviousPerformRequests(withTarget:))
```

Cancels perform requests previously registered with the [`perform(_:with:afterDelay:)`](/documentation/ObjectiveC/NSObject-swift.class/perform(_:with:afterDelay:)) instance method.

```
[`+ cancelPreviousPerformRequestsWithTarget:selector:object:`](/documentation/ObjectiveC/NSObject-swift.class/cancelPreviousPerformRequests(withTarget:selector:object:))
```

Cancels perform requests previously registered with [`perform(_:with:afterDelay:)`](/documentation/ObjectiveC/NSObject-swift.class/perform(_:with:afterDelay:)).

Forwarding Messages

```
[`- forwardingTargetForSelector:`](/documentation/ObjectiveC/NSObject-swift.class/forwardingTarget(for:))
```

Returns the object to which unrecognized messages should first be directed.

```
[`- forwardInvocation:`](/documentation/ObjectiveC/NSObject-swift.class/forwardInvocation:)
```

Overridden by subclasses to forward messages to other objects.

Dynamically Resolving Methods

```
[`+ resolveClassMethod:`](/documentation/ObjectiveC/NSObject-swift.class/resolveClassMethod(_:))
```

Dynamically provides an implementation for a given selector for a class method.

```
[`+ resolveInstanceMethod:`](/documentation/ObjectiveC/NSObject-swift.class/resolveInstanceMethod(_:))
```

Dynamically provides an implementation for a given selector for an instance method.

Handling Errors

```
[`- doesNotRecognizeSelector:`](/documentation/ObjectiveC/NSObject-swift.class/
```

`doesNotRecognizeSelector(_:))`

Handles messages the receiver doesn't recognize.

Archiving

`[`- awakeAfterUsingCoder:`](/documentation/ObjectiveC/NSObject-swift.class/awakeAfter(using:))`

Overridden by subclasses to substitute another object in place of the object that was decoded and subsequently received this message.

`[`classForArchiver`](/documentation/ObjectiveC/NSObject-swift.class/classForArchiver)`

The class to substitute for the receiver's own class during archiving.

`[`classForCoder`](/documentation/ObjectiveC/NSObject-swift.class/classForCoder)`

Overridden by subclasses to substitute a class other than its own during coding.

`[`classForKeyedArchiver`](/documentation/ObjectiveC/NSObject-swift.class/classForKeyedArchiver)`

Subclasses to substitute a new class for instances during keyed archiving.

`[`+ classFallbacksForKeyedArchiver`](/documentation/ObjectiveC/NSObject-swift.class/classFallbacksForKeyedArchiver())`

Overridden to return the names of classes that can be used to decode objects if their class is unavailable.

`[`+ classForKeyedUnarchiver`](/documentation/ObjectiveC/NSObject-swift.class/classForKeyedUnarchiver())`

Overridden by subclasses to substitute a new class during keyed unarchiving.

`[`classForPortCoder`](/documentation/ObjectiveC/NSObject-swift.class/classForPortCoder)`

Class to substitute for the receiver in distribution encoding.

`[`- replacementObjectForArchiver:`](/documentation/ObjectiveC/NSObject-swift.class/replacementObject(for:)-8ih2x)`

Overridden by subclasses to substitute another object for itself during archiving.

`[`- replacementObjectForCoder:`](/documentation/ObjectiveC/NSObject-swift.class/replacementObject(for:)-2l8ox)`

Overridden by subclasses to substitute another object for itself during encoding.

`[`- replacementObjectForKeyedArchiver:`](/documentation/ObjectiveC/NSObject-swift.class/replacementObject(for:)-60vwc)`

Overridden by subclasses to substitute another object for itself during keyed archiving.

`[`+ setVersion:`](/documentation/ObjectiveC/NSObject-swift.class/setVersion(_:))`

Sets the receiver's version number.

```
[`+ version`](/documentation/ObjectiveC/NSObject-swift.class/version())
```

Returns the version number assigned to the class.

Working with Class Descriptions

```
[`attributeKeys`](/documentation/ObjectiveC/NSObject-swift.class/attributeKeys)
```

An array of `NSString` objects containing the names of immutable values that instances of the receiver's class contain.

```
[`classDescription`](/documentation/ObjectiveC/NSObject-swift.class/classDescription)
```

An object containing information about the attributes and relationships of the receiver's class.

```
[`- inverseForRelationshipKey:`](/documentation/ObjectiveC/NSObject-swift.class/inverse(forRelationshipKey:))
```

For a given key that defines the name of the relationship from the receiver's class to another class, returns the name of the relationship from the other class to the receiver's class.

```
[`toManyRelationshipKeys`](/documentation/ObjectiveC/NSObject-swift.class/toManyRelationshipKeys)
```

An array containing the keys for the to-many relationship properties of the receiver.

```
[`toOneRelationshipKeys`](/documentation/ObjectiveC/NSObject-swift.class/toOneRelationshipKeys)
```

The keys for the to-one relationship properties of the receiver, if any.

Improving Accessibility

```
<doc://com.apple.documentation/documentation/UIKit/uiaccessibility-protocol>
```

```
<doc://com.apple.documentation/documentation/UIKit/uiaccessibilitycontainer>
```

```
[UIAccessibilityAction](/documentation/ObjectiveC/uiaccessibilityaction)
```

A set of methods that accessibility elements can use to support specific actions.

```
[UIAccessibilityFocus](/documentation/ObjectiveC/uiaccessibilityfocus)
```

An informal protocol that provides a way to determine whether an assistive app, such as VoiceOver, has focus on an accessible element.

```
[UIAccessibilityDragging](/documentation/ObjectiveC/uiaccessibilitydragging)
```

A pair of properties to allow you to fine-tune how drags and drops are exposed to assistive technologies.

Improving browser accessibility

```
[`- browserAccessibilityAttributedValueInRange:`](/documentation/ObjectiveC/NSObject-swift.class/browserAccessibilityAttributedValueInRange:)
```

`bject-swift.class/browserAccessibilityAttributedValue(in:))`

Returns the value for this element within the given range, as an attributed string.

`[`- browserAccessibilityDeleteTextAtCursor:`](/documentation/ObjectiveC/NSObject-
t-swift.class/browserAccessibilityDeleteTextAtCursor(numberOfCharacters:))`

Deletes text from the element at the current cursor position.

`[`- browserAccessibilityInsertTextAtCursor:`](/documentation/ObjectiveC/NSObject-
t-swift.class/browserAccessibilityInsertTextAtCursor(text:))`

Inserts text into the element at the current cursor position.

`[`- browserAccessibilitySelectedTextRange`](/documentation/ObjectiveC/NSObject-
swift.class/browserAccessibilitySelectedTextRange())`

Returns the range of selected text in the element.

`[`- browserAccessibilitySetSelectedTextRange:`](/documentation/ObjectiveC/NSObject-
t-swift.class/browserAccessibilitySetSelectedTextRange(_:))`

Updates the element's selected text.

`[`- browserAccessibilityValueInRange:`](/documentation/ObjectiveC/NSObject-swif
t.class/browserAccessibilityValue(in:))`

Returns this element's value in the given range.

`[`browserAccessibilityContainerType`](/documentation/ObjectiveC/NSObject-swift.c
lass/browserAccessibilityContainerType)`

The kind of container that contains this element.

`[`browserAccessibilityCurrentStatus`](/documentation/ObjectiveC/NSObject-swift.c
lass/browserAccessibilityCurrentStatus)`

A string that's the element's value for aria-current.

`[`browserAccessibilityHasDOMFocus`](/documentation/ObjectiveC/NSObject-swift.cla
ss/browserAccessibilityHasDOMFocus)`

A Boolean value that indicates whether the element has native focus in the browser Document Object Model.

`[`browserAccessibilityIsRequired`](/documentation/ObjectiveC/NSObject-swift.clas
s/browserAccessibilityIsRequired)`

A Boolean value that's the element's value for aria-required.

`[`browserAccessibilityPressedState`](/documentation/ObjectiveC/NSObject-swift.cl
ass/browserAccessibilityPressedState)`

The element's value for aria-pressed.

`[`browserAccessibilityRoleDescription`](/documentation/ObjectiveC/NSObject-swift
.class/browserAccessibilityRoleDescription)`

A string that describes the element's role for assistive technologies.

`[`browserAccessibilitySortDirection`](/documentation/ObjectiveC/NSObject-swift.c`

lass/browserAccessibilitySortDirection)

A string that's the element's value for aria-sort.

Scripting

[`classCode`](/documentation/ObjectiveC/NSObject-swift.class/classCode)

The receiver's Apple event type code, as stored in the `NSScriptClassDescription` object for the object's class.

[`className`](/documentation/ObjectiveC/NSObject-swift.class/className)

A string containing the name of the class.

[`- copyScriptingValue(forKey:withProperties:`](/documentation/ObjectiveC/NSObject-swift.class/copyScriptingValue(_:forKey:withProperties:))

Creates and returns one or more scripting objects to be inserted into the specified relationship by copying the passed-in value and setting the properties in the copied object or objects.

[`- newScriptingObjectOfClass:forValueForKey:withContentsValue:properties:`](/documentation/ObjectiveC/NSObject-swift.class/newScriptingObject(of:forValueForKey:withContentsValue:properties:))

Creates and returns an instance of a scriptable class, setting its contents and properties, for insertion into the relationship identified by the key.

[`scriptingProperties`](/documentation/ObjectiveC/NSObject-swift.class/scriptingProperties)

An `NSString`-keyed dictionary of the receiver's scriptable properties.

[`- scriptingValueForSpecifier:`](/documentation/ObjectiveC/NSObject-swift.class/scriptingValue(for:))

Given an object specifier, returns the specified object or objects in the receiving container.

Integrating with Combine

[`KeyValueObservingPublisher`](/documentation/ObjectiveC/NSObject-swift.class/KeyValueObservingPublisher)

A Combine publisher that produces a new element whenever the observed value changes.

Key-Value Observing

[`NSKeyValueObserving`](/documentation/ObjectiveC/nskeyvalueobserving)

An informal protocol that objects adopt to be notified of changes to the specified properties of other objects.

Key-Value Coding

[`NSKeyValueBindingCreation`](/documentation/ObjectiveC/nskeyvaluebindingcreation)

A set of methods that you can use to create and remove bindings between view objects and controllers, or between controllers and model objects.

[\[NSKeyValueCoding\]\(/documentation/ObjectiveC/nskeyvaluecoding\)](#)

A mechanism by which you can access the properties of an object indirectly by name or key.

[\[NSScriptKeyValueCoding\]\(/documentation/ObjectiveC/nsscriptkeyvaluecoding\)](#)

A collection of methods that provide additional capabilities for working with key-value coding.

[\[NSScriptKeyValueCoding Exception Names\]\(/documentation/ObjectiveC/nsscriptkeyvaluecoding-exception-names\)](#)

Exceptions raised by key-value coding methods.

Interacting with Web Plug-ins

[\[WebPlugInContainer\]\(/documentation/ObjectiveC/webplugincontainer\)](#)

``WebPlugInContainer`` is an informal protocol that enables a plug-in to send messages to the application.

[\[WebPlugIn\]\(/documentation/ObjectiveC/webplugin\)](#)

The ``WebPlugIn`` informal protocol defines methods that enable interaction between an application using the WebKit framework and any WebKit-based plug-ins it may use.

Implementing Web Scripting

[\[WebScripting\]\(/documentation/ObjectiveC/webscripting\)](#)

``WebScripting`` is an informal protocol that defines methods that classes can implement to export their interfaces to a WebScript environment such as JavaScript.

Supporting Cocoa Scripting

[\[NSScriptingComparisonMethods\]\(/documentation/ObjectiveC/nsscriptingcomparisonmethods\)](#)

A collection of methods useful for comparing script objects.

Customizing accessibility

[\[`accessibilityElements`\]\(/documentation/ObjectiveC/NSObject-swift.class/accessibilityElements\)](#)

An array of features of an object that assistive technologies can access.

Deprecated

Avoid using deprecated classes and protocols in your apps.

[\[Deprecated Symbols\]\(/documentation/ObjectiveC/deprecated-symbols\)](#)

Review symbols that are no longer supported and find the replacements to use instead.

Relationships

Conforms To

[`Hashable`](/documentation/Swift/Hashable)

[`CustomDebugStringConvertible`](/documentation/Swift/CustomDebugStringConvertible)

[`NSObjectProtocol`](/documentation/ObjectiveC/NSObjectProtocol)

[`Equatable`](/documentation/Swift/Equatable)

[`Copyable`](/documentation/Swift/Copyable)

[`CustomStringConvertible`](/documentation/Swift/CustomStringConvertible)

[`Escapable`](/documentation/Swift/Escapable)

[`CVarArg`](/documentation/Swift/CVarArg)

Copyright © 2026 Apple Inc. All rights reserved. | [Terms of Use](https://www.apple.com/legal/internet-services/terms/site.html) | [Privacy Policy](https://www.apple.com/privacy/privacy-policy)